

What is Cloud?

- Before cloud came into the picture we had on-prem infrastructure to host our application or website.
- Difficult to auto-scale our application.
- Maintenance.
- Cost may be high in few cases.
- Cloud provides huge space online(collection of servers).
- Buy servers on demand.(Cost effective)

Cloud Computing

Types of Cloud Computing

- Infrastructure as a Service (IaaS)
 - Provide building blocks for cloud IT
 - Provides networking, computers, data storage space
 - Highest level of flexibility
 - Easy parallel with traditional on-premises IT
- Platform as a Service (PaaS)
 - Removes the need for your organization to manage the underlying infrastructure
 - Focus on the deployment and management of your applications
- Software as a Service (SaaS)
 - Completed product that is run and managed by the service provider

Cloud service and Types

- **SaaS** – Software as a service

SaaS stands for Software as a Service. It's a software distribution model where applications are hosted by a third-party provider and made available to customers over the internet.

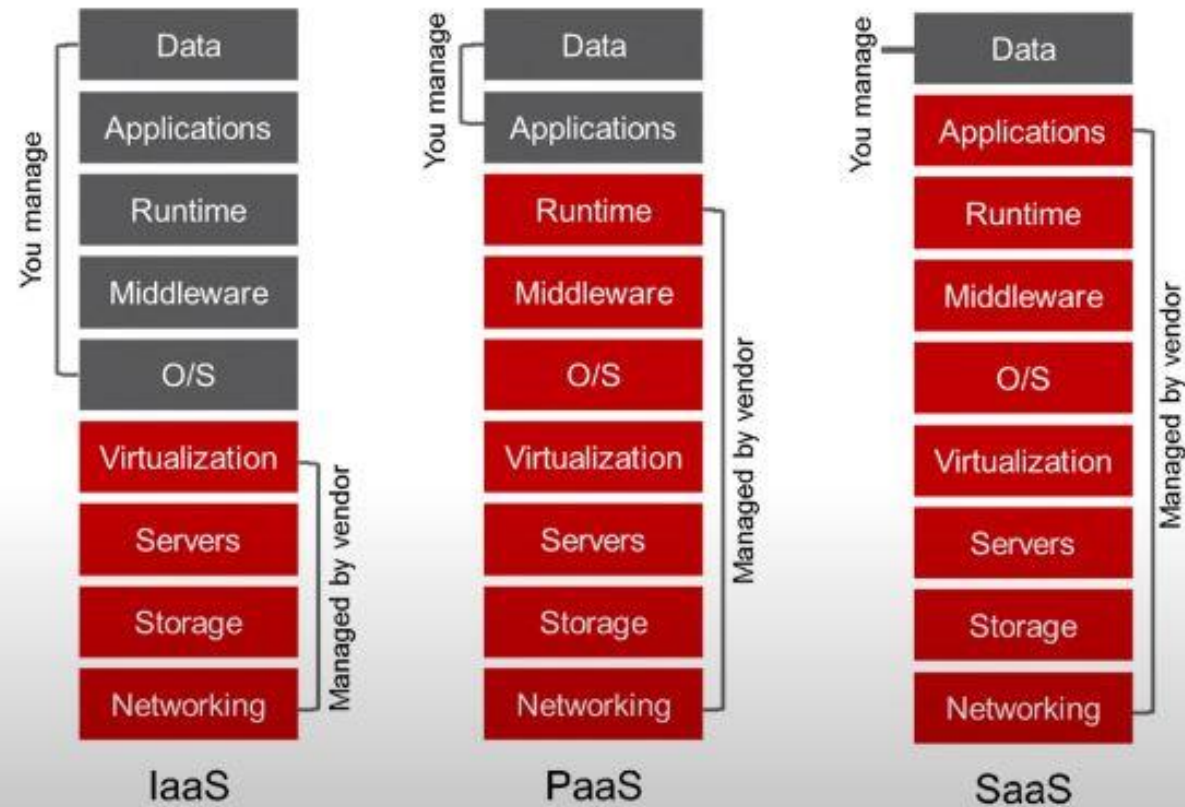
- **PaaS** –Platform as a service

Platform as a Service, is like renting a ready-to-use development platform in the cloud. Instead of worrying about setting up servers and software, you can focus on building and running your applications. It's a simpler way for developers to create and manage their software

- **IaaS** –Infrastructure as a service

IaaS stands for Infrastructure as a Service. It's like renting the basic building blocks of computing infrastructure—such as virtual machines, storage, and networking—from a cloud provider. With IaaS, you don't have to buy and maintain physical servers; instead, you can scale your computing resources up or down as needed, paying only for what you use. It's a flexible and cost-effective way to build and manage your own IT infrastructure in the cloud.

Types Of Cloud Services



AWS Region

- Geographically separated area where AWS has set its resources.
- Each region is independent of each other.
- 36 Launched Regions
- In India- Mumbai and Hyderabad
- Each region has multiple, isolated and physically separated Az's.
- **Data Center:** Think of it as a physical place, like a building or a campus, where all the servers, storage, and networking equipment are kept.
- **Region:** Yes, it's a location, but it's a broader term. A region can encompass multiple data centers and is usually a larger geographic area, like a city or a country. It's where all those data centers are situated, providing redundancy and ensuring services are available across different places within that region.

Availability Zones

- Availability Zones are a grouping of data centers within a region
- Connected through low latency links.
- In each region at least two AZ's for fault tolerance.

Edge Locations

- Small Datacentres where end user access services located at AWS.
- Frequently used data will be stored as cache, helps to access faster.
- AWS will not disclose where the AZ's and edge locations.

Identity and Access Management ---IAM

- Provides access to accounts services where we can manage User, Roles, Groups & Policy password policy.
- It applies globally to all AWS regions.
- IAM is global service , where as others are region specific

We communicate to aws account using AWS CLI command (AWS CLI to be installed)

```
aws iam list-users
```

```
aws iam list-users --output table
```

```
aws iam list-users --output text
```

```
aws iam list-users --output json
```

```
aws ec2 describe-key-pair
```

Users: we create users and assign necessary permissions to them in the form of policies.

Groups: We can create groups for ex. Dev QA etc. and attach policies at the group level.

Policy:

- A policy is a set of permission
- Always explicit deny overrides explicit allow

- **Policy types:**

Identity Based Policy: Applicable on users, groups of users, and roles

- **AWS Managed policy:** An AWS managed policy is a predefined policy made and managed by AWS to assist in permissions management for AWS resources
- **Custom Managed Policy:** A custom managed policy in AWS is a user-defined policy for managing permissions on AWS resources
- **Inline Policy:** An inline policy in AWS is a policy attached directly to an individual IAM user, group, or role for managing permissions

Resource Based policy: Attach to a resource such as an Amazon S3 bucket

A resource-based policy in AWS is a JSON document attached to a resource (such as an S3 bucket or Lambda function) defining permissions for other AWS accounts or principals to access the resource.

Session based Policy: create a temporary session for a role or federated user

A session-based policy in AWS is a temporary set of permissions assigned to an IAM role assumed by an entity for a limited duration

Note:

- More than one policy can be attached to a user or a group at the same time.
- Policies can't be attached directly to resources like EC2 instance, S3 bucket etc.,

IAM Policies Structure

- Consists of
 - Version: policy language version, always include "2012-10-17"
 - Id: an identifier for the policy (optional)
 - Statement: one or more individual statements (required)
- Statements consists of
 - Sid: an identifier for the statement (optional)
 - Effect: whether the statement allows or denies access (Allow, Deny)
 - Principal: account/user/role to which this policy applied to
 - Action: list of actions this policy allows or denies
 - Resource: list of resources to which the actions applied to
 - Condition: conditions for when this policy is in effect (optional)

```
{
  "Version": "2012-10-17",
  "Id": "S3-Account-Permissions",
  "Statement": [
    {
      "Sid": "1",
      "Effect": "Allow",
      "Principal": {
        "AWS": ["arn:aws:iam::123456789012:root"]
      },
      "Action": [
        "s3:GetObject",
        "s3:PutObject"
      ],
      "Resource": ["arn:aws:s3:::mybucket/*"]
    }
  ]
}
```

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "AllowS3ReadOnly",
      "Effect": "Allow",
      "Action": [
        "s3:GetObject",
        "s3:ListBucket"
      ],
      "Resource": [
        "arn:aws:s3:::example-bucket",
        "arn:aws:s3:::example-bucket/*"
      ]
    },
    {
      "Sid": "DenyDeleteObject",
      "Effect": "Deny",
      "Action": "s3:DeleteObject",
      "Resource": "arn:aws:s3:::example-bucket/*"
    }
  ]
}
```

Roles:

- Roles comes between services like ec2 wants to access S3 or non-AWS user (hybrid account) should access AWS Resources.
 - Policies can't be attached to aws resources hence roles come into picture
 - EC2 can be attached one role at a time.
-

- **Q: Can we attach multiple policies to user group or a role?**

- Ans: Yes, we attach.

- **Q1.Can we assign multiple roles to a EC2 instance or AWS service ?**

- Ans: No, we can't. we can assign only single role to EC2 instance or any aws service

- **Q2. If an ec2 instance is not able to access s3 bucket what could be the reason ?**

- Ans: A Role needs to be attached with proper policy defined.

- **Managed Policies: You can attach up to 10 managed policies (AWS managed or customer managed) to both a user or a role.**

- **Inline Policies: You can have multiple inline policies attached to a user, role, or group.Each inline policy can be up to 2,048 characters (including JSON formatting)**

aws sts get-caller-identity ---- To get the details about the Role being used in the server

MFA ----- We use MFA to protect our AWS account (Either root or Normal user can have a maximum of 8 MFA devices assigned.)

- **Assume Role:**

- Returns a set of temporary security credentials that you can use to access AWS resources that you might not normally have access to. These temporary credentials consist of an access key ID, a secret access key, and a security token.

- **The AWS Security Token Service (STS)** is a web service that enables you to request temporary, limited-privilege credentials for AWS Identity and Access Management (IAM) users or for users that you authenticate (federated users).